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4 / 20 submitted

Q1. Arrays

CODE

```
#include <stdio.h>
int main(){
    int production[12]={5200,3000,5200,2006,2520,5630,3000,5064,8952,4200,1200,5962};
    int i;
    int totalproduction=0;
    float efficiency;
    printf("Enter the production for 12 months:\n");
    printf("Enter -1 if the plant shuts down due to equipment failure");
    for(i=0;i<12;i++){
        printf("Month %d:",i+1);
        scanf("%d",&production[i]);
        //Break if plant shuts down
        if(production[i]==-1){
            printf("Shutdown due to equipment failure");
            continue;
        }
        //Add valid production
        totalproduction+=production[i];
        //Check incentive and inspection
        if(production[i]>5000){
            printf("performance incentive:10% awarded. \n");
        }else if(production[i]<5000){
            printf("Production unit marked for inspection. \n");
        }else{
            printf("target Acheived.\n");
        }
    }
    //calculate annual efficiency
    efficiency=(float)totalproduction/(5000*12)*100;
    printf("\n total annual production =%d batteries", totalproduction);
    printf("\n Annual production efficiency= %.2f%%",efficiency);
    return 0;
}
```

OUTPUT (no output)

Q2. Arrays

CODE

```
#include <stdio.h>
int main(){
    int quality[10]={52,90,60,92,84,25,93,88,90,89};
    int delay[10]={1,2,2,6,4,5,3,9,8,7};
    int i,total=0;
    float avg,percentage;
    for(i=0;i<10;i++){
        if(quality[i]==-1)
            break;
        if(quality[i]==-2)
            continue;
        total+=quality[i];
        if(quality[i] ≥ 90&&delay[i] ≤ 2)
            printf("Excellent\n");
        else if(quality[i] ≥ 75 && quality[i] ≤ 89 && delay[i] ≤ 5)
            printf("Good\n");
        else if(quality[i] ≥ 60)
            printf("Average\n");
        else
            printf("Needs improvement\n");
    }
    avg=total/20.0;
    percentage=avg;
    printf("\n Average Quality Score=%.2f",avg);
    printf("\n Delivery percentage=%.2f%%",percentage);
    return 0;
}
```

OUTPUT (no output)

Q3. Arrays

— Not Submitted —

Q4. Arrays

— Not Submitted —

Q5. Arrays

— Not Submitted —

Q6. Arrays

CODE

```
#include <stdio.h>
int main(){
    int defects[10]={1,2,4,8,12,3,7,9,2,5};
    int i;
    float accepted,percent;
    for(i=0;i<10;i++){
        if(defects[i]==-1)
            break;
        if(defects[i]==-2)
            continue;
        percent=(defects[i]*100.0)/10000;
        accepted=10000-defects[i];
        printf("defect percentage=%.2f%%\n",percent);
        printf("Accepted chips=%d\n",accepted);
        if(percent<1)
            printf("grade A\n");
        else if(percent≤3)
            printf("Grade B\n");
        else if(percent≤5)
            printf("Grade C\n");
        else
            printf("Reject batch\n");
        printf("\n");
    }
    return 0;
}
```

OUTPUT

(no output)

Q7. Arrays

CODE

```
#include <stdio.h>
int main(){
    int used[10]={100,50,25,98,65,23,65,75,86,92};
    int i,total,extra;
    float p;
    printf("Plan(1-basic 2-standard 3-enterprise): ");
    scanf("%d",&total);
    if(total==1)
        total=100;
    else if(total ==2)
        total=500;
    else
        total=2000;
    for(i=0;i<10;i++){
        if(used[i]==-1)
            break;
        if(used[i]==-2)
            continue;
        p=used[i]*100.0/total;
        extra=(used[i]>total)?(used[i]-total)*24:0;
        printf("remaining=%d GB\n",total-used[i]);
        printf("extra charges=%d\n",extra);
        if(p>95)
            printf("warning\n");
        else
            printf("Normal\n");
    }
    return 0;
}
```

OUTPUT

(no output)

Q8. Arrays

— Not Submitted —

Q9. Arrays

— Not Submitted —

Q10. Arrays

— Not Submitted —

Q11. Arrays

— Not Submitted —

Q12. Arrays

— Not Submitted —

Q13. Arrays

— Not Submitted —

Q14. Arrays

— Not Submitted —